

Name _____

Environments and Closures Practice In Class

DAY 2

In Racket, what order are the procedures A B C D E called?

A (1 point)

```
((lambda (fun1 fun2)
```

```
  (A 99)
```

```
  (fun1 99)
```

```
  (fun2 99))
```

B

```
(C 99))
```

B (1 point)

```
(let ([myFuncs (list A B)])
```

```
  ((cadr myFuncs)
```

```
    ((car myFuncs) 7)
```

```
    (C D)))
```

Draw E&C Diagrams for the following code:

A (4 points)

```
(let ([A (lambda (x) (display 'A))]
      [B (lambda (x y) (display 'B))]
      [C (lambda (x) (display 'C))]
      [D (lambda (x) (display 'D))])
  (let ([myFuncs (list A B)])
    ((cadr myFuncs)
     ((car myFuncs) 7)
     (C D))))
```

B (4 points) <- if you finish this and get it checked off, you can leave

```
(define make-weird-counter
  (let ((supply 5)) ; think about how many times this
    let is run
      (lambda ()
        (let [(num 0)] ; how many times this let is run
          (lambda ()
            (if (zero? supply) 'supply-empty
                (begin
                  (set! num (add1 num))
                  (set! supply (sub1 supply))
                  (list supply num))))))))))

(define wc1 (make-weird-counter))
(wc1)
```